7.5 Conversion of Solids from one shape to another with no change in Volume

Conversions or Transformations becomes a common part of our daily life. For example, a gold smith melts a bar of gold to transform it to a jewel. Similarly, a kid playing with clay shapes it into different toys, a carpenter uses the wooden logs to form different house hold articles/furniture. Likewise, the conversion of solids from one shape to another is required for various purposes.

In this section we will be learning problems involving conversions of solids from one shape to another with no change in volume.

Example 7.29 A metallic sphere of radius 16 cm is melted and recast into small spheres each of radius 2 cm. How many small spheres can be obtained?

Solution Let the number of small spheres obtained be n.

Let r be the radius of each small sphere and R be the radius of metallic sphere.

Here, R = 16 cm, r = 2 cm

Now, $n\times$(Volume of a small sphere) $=$ Volume of big metallic sphere

$$n\left(\frac{4}{3}\pi r^3\right) = \frac{4}{3}\pi R^3$$

$$n\left(\frac{4}{3}\pi \times 2^3\right) = \frac{4}{3}\pi \times 16^3$$

$$8n = 4096 \Rightarrow n = 512$$

Therefore, there will be 512 small spheres.

Example 7.30 A cone of height 24 cm is made up of modeling clay. A child reshapes it in the form of a cylinder of same radius as cone. Find the height of the cylinder.

Solution Let h_1 and h_2 be the heights of a cone and cylinder respectively.

Also, let r be the raius of the cone.

Given that, height of the cone $h_1 = 24$ cm; radius of the cone and cylinder $r = 6$ cm

Since, Volume of cylinder $=$ Volume of cone

$$\pi r^2 h_2 = \frac{1}{3}\pi r^2 h_1$$

$$h_2 = \frac{1}{3}\times h_1 \Rightarrow h_2 = \frac{1}{3}\times 24 = 8$$

Therefore, height of cylinder is 8 cm

Example 7.31 A right circular cylindrical container of base radius 6 cm and height 15 cm is full of ice cream. The ice cream is to be filled in cones of height 9 cm and base radius 3 cm, having a hemispherical cap. Find the number of cones needed to empty the container.

Solution Let h and r be the height and radius of the cylinder respectively.

Given that, $h = 15$ cm, $r = 6$ cm

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